TempShift: Identifying Root Causes of Global Temperature Change

Using Multi-Source Data Integration and Regression Feature Ranking

CMPSC 445 - Applied Machine Learning
Pennsylvania State University, Abington
March 2026

1. Project Description

This project builds a regression-based pipeline to identify the strongest environmental drivers of global surface temperature change. Data is collected from four independent scientific sources, merged into a unified monthly dataset, and used to train three regression models. Feature importance techniques are then applied to rank which variables most strongly predict temperature variation.

The goal is to determine whether human-driven factors like CO_2, CH_4, and industrial emissions are more influential than natural factors like solar irradiance and volcanic activity — and to quantify how much each contributes to the observed warming trend from 1984 to 2026.

1.1 Data Sources

Source	**Data Collected**	**Resolution**	**URL**
NOAA GML	CO_2 (Mauna Loa), CH_4, N_2O	Monthly	gml.noaa.gov/ccgg/trends
NASA GISS	Global temperature anomaly	Monthly	data.giss.nasa.gov/gistemp
NOAA NCEI	Total Solar Irradiance	Monthly	ncei.noaa.gov/data/total-solar-irradiance
Our World in Data	CO_2/GHG emissions by sector	Annual	ourworldindata.org/co2-emissions

Table 1. Data sources used in this project.

2. Data Preprocessing

All four sources were downloaded programmatically, cleaned, and merged into a single unified monthly dataset spanning January 1984 through February 2026.

2.1 Temporal Alignment

The four sources come at different temporal resolutions. NOAA GML [1][2] and NASA GISS [3] both provide monthly data. NOAA NCEI [4] provides monthly solar irradiance. Our World in Data [5] provides only annual observations, so each year's values were broadcast to all 12 months before merging on a shared YYYY-MM date key.
NOAA CH_4 and N_2O files use a decimal-year format (e.g., 1983.542) instead of explicit year/month columns. Year and month were derived from the decimal year mathematically. The dataset was trimmed to January 1984, the earliest point where all sources have full coverage.

2.2 Cleaning

NOAA GML [1][2] uses sentinel values (-999.99, -9.99) to mark missing observations, which were replaced with NaN. Short gaps in CO_2 and temperature series were filled by linear interpolation (limit 6 months). CH_4 and N_2O were interpolated without a limit since they start partway through the record. OWID [5] annual columns were forward-filled after broadcasting.

2.3 Feature Engineering

Seven features were engineered from the raw measurements:

- time_since_1960: Integer months since January 1960, capturing the long-term warming trend.
- co2_growth_rate / ch4_growth_rate: Month-over-month first differences, capturing rate of change.
- co2_12mo_avg / ch4_12mo_avg: 12-month rolling means, smoothing seasonal cycles.
- tsi_anomaly: Deviation of solar proxy from its 1961-1990 baseline mean.
- volcanic_flag: Binary indicator for known major eruption windows (Agung 1963, El Chichon 1982, Pinatubo 1991).
- total_anthropogenic_co2: Sum of sector CO_2 from OWID [5] (coal, oil, gas, cement, land-use change).

2.4 Final Dataset

After cleaning and trimming, the final dataset contains 874 rows and 29 columns with zero missing values.

Date	**Temp Anomaly (C)**	**CO2 (ppm)**	**CH4 (ppb)**	**N2O (ppb)**
1984-01	0.31	344.21	1639.1	316.3
2005-06	0.63	381.18	1773.2	319.7
2026-02	1.22	430.50	1939.2	339.3

Table 2. Three example rows from the preprocessed dataset. All greenhouse gas concentrations show clear upward trends alongside rising temperature.

3. Model Development

3.1 Target Variable

The target is global temperature anomaly from NASA GISS [3] (degrees Celsius relative to the 1951-1980 baseline). However, predicting absolute temperature fails on a chronological split

because the test period (2015-2026) averages about 0.52C warmer than the training period (1984-2015), causing systematic underprediction regardless of model quality.
To fix this, models are trained to predict the month-over-month temperature change (first difference). This removes the distribution shift between train and test. Absolute temperature is then reconstructed by adding each predicted change to the previous month's known temperature, and reconstruction R-squared is reported as the primary metric.

3.2 Train/Test Split

Chronological 70/30 split. First 603 months (October 1984 through April 2015) are training; the remaining 259 months (May 2015 through February 2026) are test. No random shuffling — shuffling would let the model see future observations during training.

3.3 Models

- Linear Regression: Inputs standardized with StandardScaler. Standardized coefficients used as feature importance.
- Random Forest: 300 trees, max depth 6, min samples per leaf 5. Built-in feature_importances_ used for ranking.
- Gradient Boosted Trees (XGBoost): 300 rounds, learning rate 0.03, max depth 3, L1/L2 regularization.

3.4 Performance

Model	**Diff R2**	**Abs R2 (Reconstr.)**	**MAE (C)**	**RMSE (C)**
Linear Regression	-2.304	0.414	0.125	0.149
Random Forest	-0.410	0.750	0.063	0.098
Gradient Boost (XGBoost)	-0.582	0.720	0.068	0.103

Table 3. Model performance on the test set (2015-2026). Diff R2 measures prediction of month-to-month temperature change. Abs R2 measures accuracy of reconstructed absolute temperature anomaly.

Negative Diff R2 is expected — month-to-month temperature changes are dominated by natural variability and are difficult to predict from concentration data alone. The meaningful metric is reconstructed absolute R2: Random Forest achieves 0.75 and XGBoost achieves 0.72, meaning the feature set explains roughly 72-75% of the variance in absolute temperature anomaly over the test period.

4. Root-Cause Identification via Feature Ranking

4.1 Method

Feature importances from all three models were normalized to [0,1] and averaged into a consensus score. For linear regression, absolute standardized coefficients were used. For Random Forest and XGBoost, built-in feature_importances_ were used. Averaging across models reduces the effect of any single model's tendencies.

4.2 Consensus Ranking

Rank	Feature	Consensus Score	Interpretation
1	temp_lag1 (autoregressive)	0.782	Last month's temperature - temporal persistence
2	CO2 Deseasonalized	0.584	Strongest true climate driver
3	CO2 12-mo Average	0.490	Smoothed CO2 trend signal
4	CO2 Growth Rate	0.450	Month-over-month CO2 acceleration
5	Total GHG (OWID)	0.426	Aggregate global emissions
6	CH4 Growth Rate	0.408	Methane acceleration
7	CH4 (ppb)	0.393	Methane concentration level
11	Time Since 1960	0.302	Background secular warming trend
16	Solar Irradiance Anomaly	0.217	Solar cycle variability
17	Volcanic Flag	0.004	Eruption cooling signal

Table 4. Consensus feature importance ranking. temp_lag1 is an autoregressive artifact. Ranks 2-17 represent actual climate drivers.

4.3 Human vs. Natural Drivers

Anthropogenic Factors

CO2 concentration and its derived features dominate ranks 2-5. The data from NOAA GML [1][2] shows CO2 rising from 344 ppm in 1984 to 430 ppm in 2026 — an increase of 86 ppm — while NASA GISS [3] temperature anomaly rose from roughly 0.3C to over 1.4C over the same

period. The scatter plot between CO2 and temperature shows a tight positive relationship with slope 0.0115 C/ppm, consistent with this.
Total GHG emissions from Our World in Data [5] appear at rank 5, and CH4 appears in ranks 6-7. The time_since_1960 feature at rank 11 also reflects the anthropogenic trend since human emissions have grown monotonically over this period.

Natural Factors

Solar irradiance anomaly (rank 16, score 0.217) derived from NOAA NCEI [4] shows a modest contribution, consistent with the solar cycle having a small but measurable effect on decadal-scale temperature. The volcanic activity flag (rank 17, score 0.004) has near-zero importance — the major eruptions in this dataset affect only a small fraction of months and are overwhelmed by the long-term warming trend.

5. Visualization

5.1 Time-Series

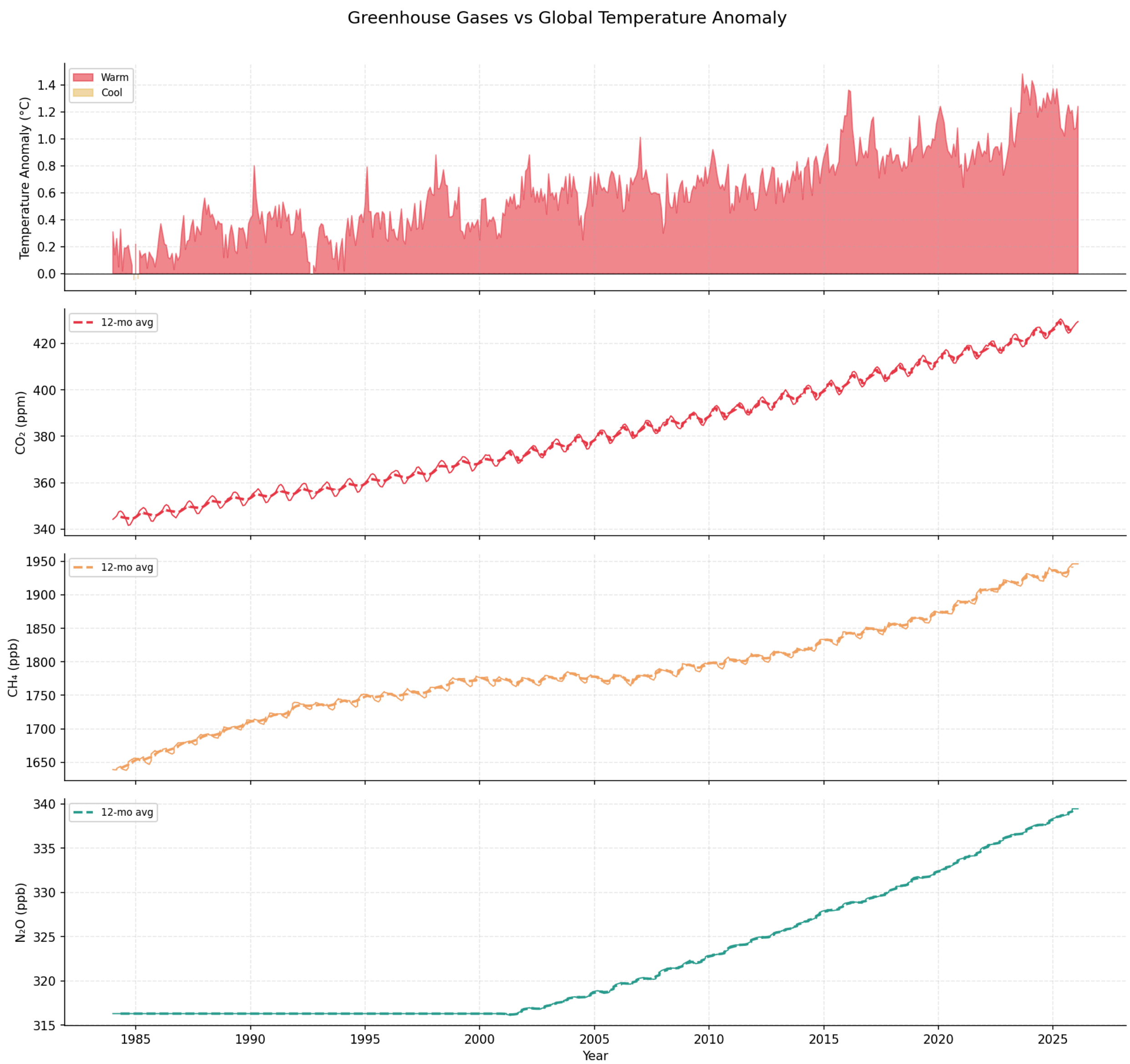

Figure 1. Monthly time series (1984-2026) of global temperature anomaly (NASA GISS [3]), CO2 (NOAA GML [1]), CH4 (NOAA GML [2]), and N2O (NOAA GML [2]). All four series trend upward.

5.2 Feature Importance

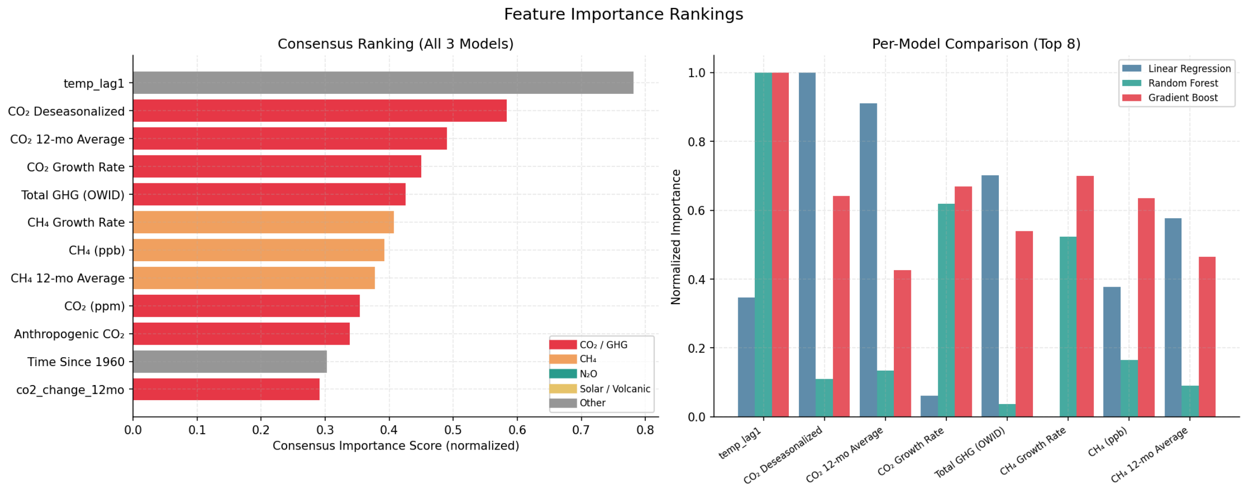

Figure 2. Consensus feature importance ranking across all three models. CO2-related features consistently dominate. Solar and volcanic signals rank lowest.

5.3 Scatter Plots

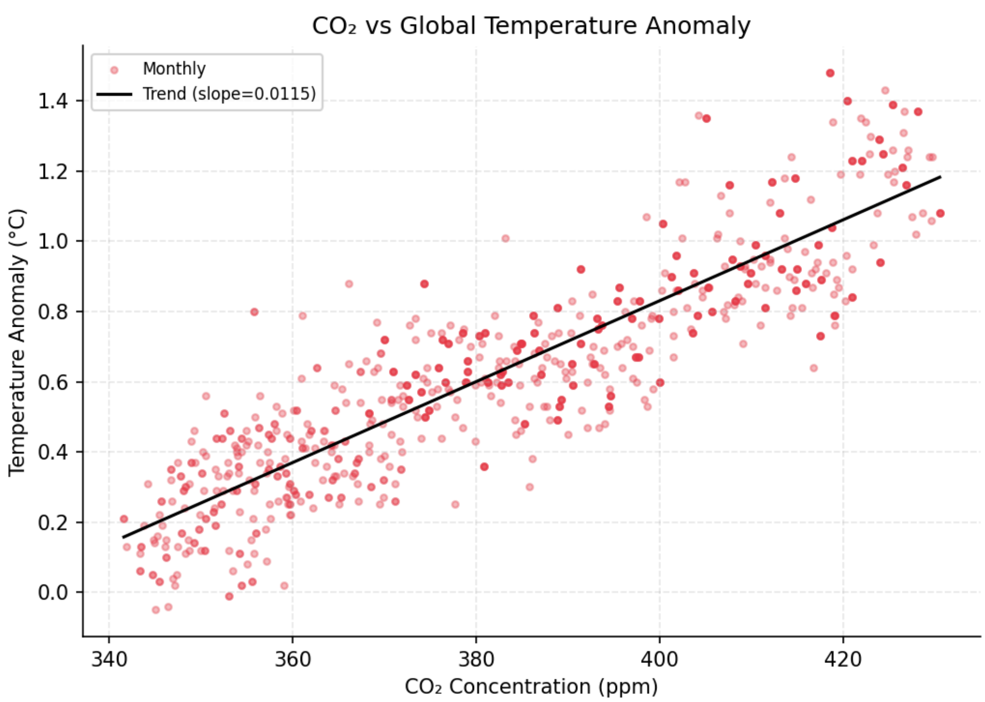

Figure 3. CO2 concentration (NOAA GML [1]) vs. global temperature anomaly (NASA GISS [3]). Strong positive relationship, slope = 0.0115 C/ppm.

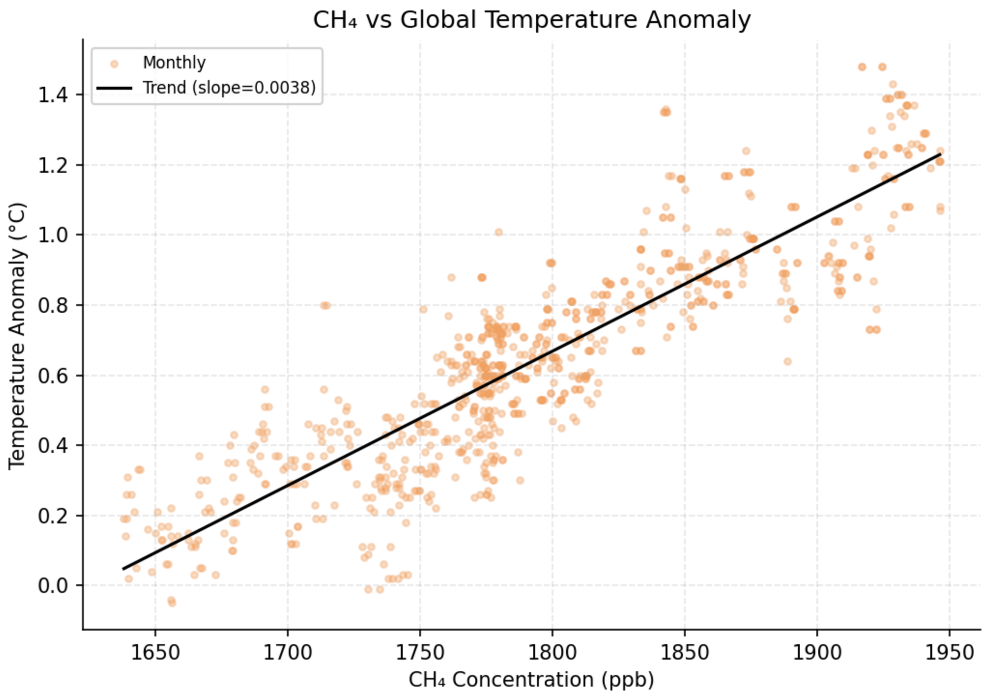

Figure 4. CH4 concentration (NOAA GML [2]) vs. global temperature anomaly (NASA GISS [3]). Similarly strong positive trend, slope = 0.0038 C/ppb.

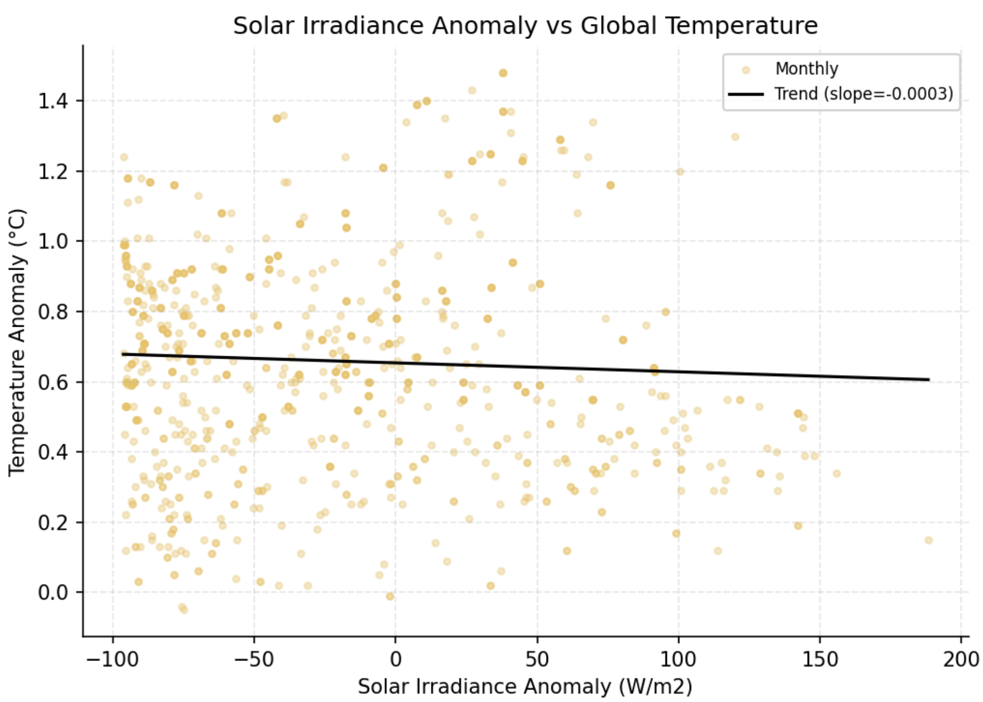

Figure 5. Solar irradiance anomaly (NOAA NCEI [4]) vs. global temperature anomaly. Near-zero slope confirms solar variability is not a primary driver of the observed warming.

5.4 Model Performance and Residuals

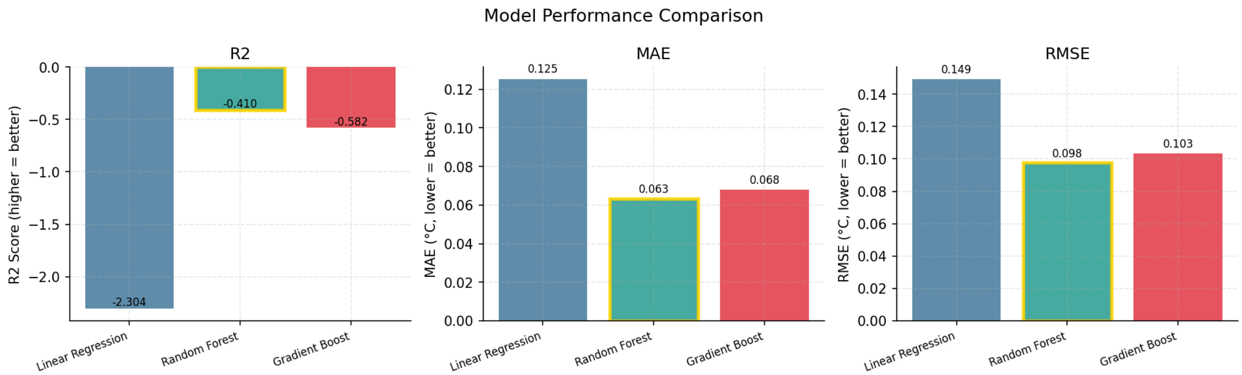

Figure 6. Model performance comparison. Random Forest achieves the lowest MAE (0.063 C) and RMSE (0.098 C).

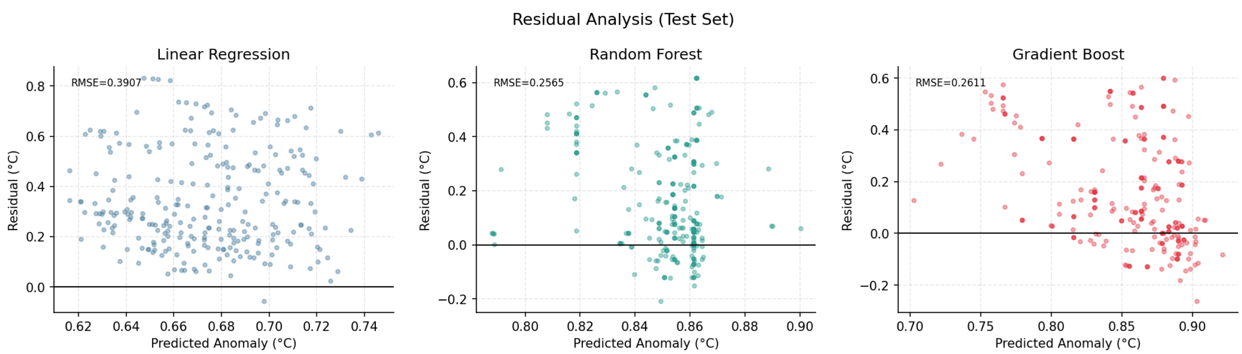

Figure 7. Residual plots on the test set. Linear Regression shows systematic upward bias. Random Forest and Gradient Boost residuals are more centered around zero.

6. Discussion and Conclusion

6.1 Key Findings

The consensus ranking consistently identifies CO2 and its derived features as the strongest predictors of temperature change. The CO2 record from NOAA GML [1] shows an 86 ppm increase between 1984 and 2026 that corresponds closely to the roughly 1.0C of warming recorded by NASA GISS [3] over the same period. CH4 from NOAA GML [2] is the second strongest human-driven signal. Solar irradiance from NOAA NCEI [4] and volcanic activity both rank at the bottom, confirming that natural forcings are secondary over this period.

6.2 Challenges

The NOAA NCEI [4] solar irradiance files are blocked by robots.txt and their newer format is netCDF rather than CSV. The pipeline falls back to SILSO sunspot numbers as a solar activity proxy when direct TSI data is unavailable, which introduces some uncertainty into the solar forcing estimates.

NOAA GML [2] CH4 and N2O files use a decimal-year format rather than explicit year/month columns, requiring custom parsing. Our World in Data [5] annual data required broadcasting to monthly resolution before merging, which assumes within-year values are constant.
The chronological train/test split creates a distribution shift — the test period is about 0.52C warmer on average than the training period — which causes all models to underpredict when targeting absolute temperature. Switching to a first-difference target resolved this.

6.3 Limitations

Feature importance identifies statistical association, not causation. All greenhouse gas concentrations are strongly correlated with each other and with time, creating multicollinearity. The high ranking of time_since_1960 is a direct result of this — it summarizes the same trend captured collectively by the GHG features. Regression models also cannot account for feedback mechanisms, ocean heat uptake, or aerosol effects that physically-based climate models handle explicitly.

6.4 Ethical Considerations

All data originates from government-operated scientific institutions and open-data organizations. The SILSO sunspot proxy substitution for real TSI data is a limitation that introduces uncertainty into solar forcing estimates and should be clearly noted whenever these results are cited.
This project identifies statistical associations between greenhouse gas concentrations and temperature. The results are consistent with the broader scientific literature but this analysis alone should not be cited as independent evidence of causation — regression on observational data cannot replace the formal attribution methods used in peer-reviewed climate science.

6.5 Conclusion

The pipeline successfully merges data from four sources, engineers meaningful features, and trains three regression models to predict global temperature change. CO2 and other greenhouse gas concentrations are the dominant predictors of observed warming. Solar variability and volcanic activity play minor roles. Random Forest achieves the best absolute reconstruction performance (R2 = 0.75, MAE = 0.063 C). These findings are consistent with what the source datasets from NOAA GML [1][2], NASA GISS [3], NOAA NCEI [4], and Our World in Data [5] collectively show: a strong and accelerating relationship between rising greenhouse gas concentrations and global surface temperature.

7. AI Usage Statement

Claude (Anthropic) was used as an AI assistant during this project in the following ways:
Code scaffolding: The initial project structure and module organization were drafted with Claude's assistance.
Debugging: Bugs were diagnosed and fixed with Claude's help, including pandas 2.x API changes, NOAA file format inconsistencies in CH_4/N_2O decimal-year parsing, and the chronological distribution shift causing negative R-squared values.
Conceptual guidance: Claude explained the distribution shift problem and suggested the first-difference target approach.

All scientific interpretation, report writing, and final methodology decisions were made by the student. The code was executed and tested on a local machine running Asahi Fedora Linux.

8. References

[1] NOAA Global Monitoring Laboratory - CO2 Trends:
https://gml.noaa.gov/ccgg/trends/
[2] NOAA Global Monitoring Laboratory:
https://www.gml.noaa.gov/
[3] NASA GISS Surface Temperature Analysis (GISTEMP v4):
https://data.giss.nasa.gov/gistemp/
[4] NOAA NCEI Total Solar Irradiance Data Access:
https://www.ncei.noaa.gov/data/total-solar-irradiance/access/
[5] Our World in Data - CO2 and Greenhouse Gas Emissions:
https://ourworldindata.org/co2-emissions